APPENDIX A

Physical Property Data

A.1 CRITICAL CONSTANTS, ACENTRIC FACTORS, AND ANTOINE COEFFICIENTS:[1]

The Antoine equation is of the form: $\ln(P^{\text{sat}}\,[\text{bar}]) = A - \dfrac{B}{T[K] + C}$

TABLE A.1.1 Organic compounds

Formula	Name	$MW_{[\text{g/mol}]}$	T_c [K]	P_c [bar]	ω	A	B	C	$T_{\min}$	T_{mix}
CH_2O	Formaldehyde	30.026	408	65.86	0.253	9.8573	2204.13	−30.15	185	271
CH_4	Methane	16.042	190.6	46.00	0.008	8.6041	897.84	−7.16	93	120
CH_4O	Methanol	32.042	512.6	80.96	0.559	11.9673	3626.55	−34.29	257	364
C_2H_4	Acetylene	26.038	308.3	61.40	0.184	9.7279	1637.14	−19.77	194	202
C_2H_3N	Acetonitrile	41.052	548	48.33	0.321	9.6672	2945.47	−49.15	260	390
C_2H_4	Ethylene	28.053	282.4	50.36	0.085	8.9166	1347.01	−18.15	120	182
C_2H_4O	Acetaldehyde	44.053	461	55.73	0.303	9.6279	2465.15	−37.15	210	320
C_2H_4O	Ethylene oxide	44.053	469	71.94	0.200	10.1198	2567.61	−29.01	300	310
$C_2H_4O_2$	Acetic acid	60.052	594.4	57.86	0.454	10.1878	3405.57	−56.34	290	430
C_2H_6	Ethane	30.069	305.4	48.74	0.099	9.0435	1511.42	−17.16	130	199
C_2H_6O	Ethanol	46.068	516.2	63.83	0.635	12.2917	3803.98	−41.68	270	369
C_3H_6	Propylene	42.080	365.0	46.20	0.148	9.0825	1807.53	−26.15	160	240
C_3H_6O	Acetone	58.079	508.1	47.01	0.309	10.0311	2940.46	−35.93	241	350
C_3H_8	Propane	44.096	370.0	42.44	0.152	9.1058	1872.46	−25.16	164	249
C_3H_8O	1-Propanol	60.095	536.7	51.68	0.624	10.9237	3166.38	−80.15	285	400
C_4H_6	1,3-Butadiene	54.090	425	43.27	0.195	9.1525	2142.66	−34.30	215	290
C_4H_8	*cis*-2-Butene	56.106	435.6	42.05	0.202	9.1969	2210.71	−36.15	200	305
C_4H_8	*trans*-2-Butene	56.106	428.6	41.04	0.214	9.1975	2212.32	−33.15	200	300
$C_4H_8O_2$	Ethyl acetate	88.105	523.2	38.30	0.363	9.5314	2790.50	−57.15	260	385
C_4H_{10}	*n*-Butane	58.122	425.2	37.90	0.193	9.0580	2154.90	−34.42	195	290
C_4H_{10}	Isobutane	58.122	408.1	36.48	0.176	8.9179	2032.76	−33.15	187	280
$C_4H_{10}O$	*n*-Butanol	74.122	562.9	44.18	0.590	10.5958	3137.02	−94.43	288	404
C_5H_{10}	1-Pentene	70.133	464.7	40.53	0.245	9.1444	2405.96	−39.63	220	325
C_5H_{12}	*n*-Pentane	72.149	469.6	33.74	0.251	9.2131	2477.07	−39.94	220	330
C_6H_6	Benzene	78.112	562.1	48.94	0.212	9.2806	2788.51	−52.36	280	377
C_6H_6O	Phenol	94.111	694.2	61.30	0.440	9.8077	3490.89	−98.59	345	481
C_6H_7N	Aniline	93.127	699	53.09	0.382	10.0546	3857.52	−73.15	340	500
C_6H_{12}	Cyclohexane	84.159	553.4	40.73	0.213	9.1325	2766.63	−50.50	280	380
C_6H_{12}	1-Hexene	84.159	504.0	31.71	0.285	9.1887	2654.81	−47.30	240	360
C_6H_{14}	*n*-Hexane	86.175	507.4	29.69	0.296	9.2164	2697.55	−48.78	245	370

(*Continued*)

[1] For a more complete set of compounds, consult ThermoSolver, the text software.

TABLE A.1.1 Continued

Formula	Name	$MW_{[g/mol]}$	T_c [K]	P_c [bar]	ω	A	B	C	T_{min}	T_{max}
C_7H_8	Toluene	92.138	591.7	41.14	0.257	9.3935	3096.52	−53.67	280	410
C_7H_{14}	1-Heptene	98.186	537.2	28.37	0.358	9.2692	2895.51	−53.97	265	400
C_7H_{16}	*n*-Heptane	100.202	540.2	27.36	0.351	9.2535	2911.32	−56.51	270	400
C_8H_8	Styrene	104.149	647.0	39.92	0.257	9.3991	3328.57	−63.72	305	460
C_8H_{10}	*o*-Xylene	106.165	630.2	37.29	0.314	9.4954	3395.57	−59.46	305	445
C_8H_{10}	*m*-Xylene	106.165	617.0	35.46	0.331	9.5188	3366.99	−58.04	300	440
C_8H_{10}	*p*-Xylene	106.165	616.2	35.16	0.324	9.4761	3346.65	−57.84	300	440
C_8H_{10}	Ethylbenzene	106.165	617.1	36.07	0.301	9.3993	3279.47	−59.95	300	450
C_8H_{16}	1-Octene	112.213	566.6	26.24	0.386	9.3428	3116.52	−60.39	288	420
C_8H_{18}	*n*-Octane	114.229	568.8	24.82	0.394	9.3224	3120.29	−63.63	292	425
C_9H_{20}	*n*-Nonane	128.255	594.6	23.10	0.444	9.3469	3291.45	−71.33	312	452
$C_{10}H_8$	Naphthalene	128.171	748.4	40.53	0.302	9.5224	3992.01	−71.29	360	545
$C_{10}H_{22}$	*n*-Decane	142.282	617.6	21.08	0.490	9.3912	3456.80	−78.67	330	476

TABLE A.1.2 Inorganic Compounds

Formula	Name	$MW_{[g/mol]}$	T_c [K]	P_c [bar]	ω	A	B	C	T_{min}	T_{max}
Ar	Argon	39.948	150.8	48.74	−0.004	8.6128	700.51	−5.84	81	94
BCl_3	Boron trichloride	117.169	451.95	38.71	0.148	9.0985	2242.71	−38.99	182	286
B_2H_6	Diborane	27.670	289.80	40.50	0.138	8.7074	1377.84	−22.18	118	181
Br_2	Bromine	159.808	584	103.35	0.132	9.2239	2582.32	−51.56	259	354
CCl_3F	Trichlorofluoromethane	137.367	471.2	44.08	0.188	9.2314	2401.61	−36.3	240	300
CF_4	Carbon tetrafluoride	88.004	227.6	37.39	0.191	9.4341	1244.55	−13.06	93	148
C_2F_6	Hexafluoroethane	138.012	292.8	30.42	0.255	9.1646	1559.11	−24.51	180	195
$CHCl_3$	Chloroform	119.377	536.4	54.72	0.216	9.3530	2696.79	−46.16	260	370
CO	Carbon monoxide	28.010	132.9	34.96	0.049	7.7484	530.22	−13.15	63	108
CO_2	Carbon dioxide	44.010	304.2	73.76	0.225	15.9696	3103.39	−0.16	154	204
CS_2	Carbon disulfide	76.143	552	79.03	0.115	9.3642	2690.85	−31.62	228	342
Cl_2	Chlorine	70.905	417	77.01	0.073	9.3408	1978.32	−27.01	172	264
F_2	Fluorine	37.997	144.3	52.18	0.048	9.0498	714.10	−6.00	59	91
H_2	Hydrogen	2.016	33.2	12.97	−0.22	7.0131	164.90	3.19	14	25
HBr	Hydrogen bromide	80.912	363.2	85.52	0.063	7.8485	1242.53	−47.86	184	221
HCN	Hydrogen cyanide	27.025	456.8	53.90	0.407	9.8936	2585.80	−37.15	234	330
HCl	Hydrogen chloride	36.461	324.6	83.09	0.12	9.8838	1714.25	−14.45	137	200
H_2O	Water	18.015	647.3	220.48	0.344	11.6834	3816.44	−46.13	284	441
H_2S	Hydrogen sulfide	34.082	373.2	89.37	0.100	9.4838	1768.69	−26.06	190	230
NH_3	Ammonia	17.031	405.6	112.77	0.250	10.3279	2132.50	−32.98	179	261
He	Helium-4	4.003	5.19	2.27	−0.387	5.6312	33.7329	1.79	3.7	4.3
HF	Hydrogen fluoride	20.006	461	64.85	0.372	11.0756	3404.49	15.06	206	313
Kr	Krypton	83.800	209.4	55.02	−0.002	8.6475	958.75	−8.71	113	129
N_2	Nitrogen	28.013	126.2	33.84	0.039	8.3340	588.72	−6.60	54	90
NF_3	Nitrogen trifluoride	71.002	234	45.29	0.132	8.9905	1155.69	−15.37	103	155
N_2O	Nitrous oxide	44.013	309.6	72.45	0.160	9.5069	1506.49	−25.99	144	200
NO	Nitric oxide	30.006	180	64.85	0.607	13.5112	1572.52	−4.88	95	140
NO_2	Nitrogen dioxide	46.006	431.4	101.33	0.86	13.9122	4141.29	3.65	230	320
Ne	Neon	20.180	44.4	27.56	0.00	7.3897	180.47	−2.61	24	29
O_2	Oxygen	31.999	154.6	50.46	0.021	8.7873	734.55	−6.45	63	100
PH_3	Phosphene	33.998	324.45	65.35	0.042	9.2700	1617.91	−11.07	144	186
SF_6	Sulfur hexafluoride	146.056	318.7	37.59	0.286	12.7583	2524.78	−11.16	159	220
SO_2	Sulfur dioxide	64.065	430.8	78.83	0.251	10.1478	2302.35	−35.97	195	280

TABLE A.1.2 Continued

Formula	Name	$MW_{[g/mol]}$	T_c [K]	P_c [bar]	ω	A	B	C	T_{min}	T_{max}
SO_3	Sulfur trioxide	80.064	491.0	82.07	0.41	14.2201	3995.70	−36.66	290	332
$SiCl_3H$	Trichlorosilane	135.452	479.0	41.7	0.203	9.7079	2694.02	−27.00	275	305
$SiCl_4$	Silicon tetrachloride	169.896	507.0	37.49	0.264	9.1817	2634.16	−43.15	238	364
SiF_4	Silicon tetrafluoride	104.079	259.09	37.15	0.456	16.3709	2810.45	−6.88	129	128
SiH_4	Silane	32.117	269.69	48.43	0.089	9.7222	1620.99	5.35	94	162
WF_6	Tungsten hexafluoride	297.830	444.0	43.40	0.231	10.4899	2351.42	−64.70	202	290

Sources: Mostly from R. C. Reid, J. M. Prausnitz, and T. K. Sherwood. *The Properties of Gases and Liquids*, 3rd ed. (New York: McGraw-Hill, 1977). Also from: CRC Handbook of Chemistry and Physics (Boca Raiton, FL CRC Press, (various) years); P. J. Linstrom and W. G. Mallard, Eds., **NIST Chemistry WebBook, NIST Standard Reference Database Number 69,** June 2005, National Institute of Standards and Technology, Gaithersburg MD, 20899 (http://webbook.nist.gov/chemistry/fluid).; C. L. Yaws, *Handbook of Vapor Pressure* (vol. 4) (Houston: Gulf Publishing, 1995).

▶ A.2 HEAT CAPACITY DATA

$$\frac{c_p}{R} = A + BT + CT^2 + DT^{-2} + ET^3 \text{ with } T \text{ in } [\text{K}]$$

TABLE A.2.1 Heat Capacity of Ideal gases: Organic Compounds

Formula	Name	A	$B \times 10^3$	$C \times 10^6$	$D \times 10^{-5}$	$E \times 10^9$	T_{min}	T_{max}	Source
CH_2O	Formaldehyde	2.264	7.022	−1.877			298	1500	1
CH_4	Methane	1.702	9.081	−2.164			298	1500	1
CH_4O	Methanol	2.211	12.216	−3.45			298	1500	1
C_2H_2	Acetylene	6.132	1.952		−1.299		298	1500	1
C_2H_4	Ethylene	1.424	14.394	−4.392			298	1500	1
C_2H_4O	Acetaldehyde	1.693	17.978	−6.158			298	1000	1
C_2H_4O	Ethylene oxide	−0.385	23.463	−9.296			298	1000	1
C_2H_6	Ethane	1.131	19.225	−5.561			298	1500	1
C_2H_6O	Ethanol	3.518	20.001	−6.002			298	1500	1
C_3H_6	Propylene	1.637	22.706	−6.915			298	1500	1
C_3H_8	Propane	1.213	28.785	−8.824			298	1500	1
C_4H_6	l.3-Butadiene	2.734	26.786	−8.882			298	1500	1
C_4H_8	1-Butene	1.967	31.63	−9.873			298	1500	1
C_4H_{10}	*n*-Butane	1.935	36.915	−11.402			298	1500	1
C_4H_{10}	Isobutane	1.677	37.853	−11.945			298	1500	1
C_5H_{10}	1-Pentene	2.691	39.753	−12.447			298	1500	1
C_5H_{12}	*n*-Pentane	2.464	45.351	−14.111			298	1500	1
C_6H_6	Benzene	−0.206	39.064	−13.301			298	1500	1
C_6H_{12}	Cyclohexane	−3.876	63.249	−20.928			298	1500	1
C_6H_{12}	1-Hexene	3.220	48.189	−15.157			298	1500	1
C_6H_{14}	*n*-Hexane	3.025	53.722	−16.791			298	1500	1
C_7H_8	Toluene	0.290	47.052	−15.716			298	1500	1
C_7H_{14}	1-Heptene	3.768	56.588	−17.847			298	1500	1
C_7H_{16}	*n*-Heptane	3.570	62.127	−19.468			298	1500	1
C_8H_8	Styrene	2.050	50.192	−16.662			298	1500	1
C_8H_{10}	Ethylbenzene	1.124	55.38	−18.476			298	1500	1
C_8H_{16}	1-Octene	4.324	64.96	−20.521			298	1500	1
C_8H_{18}	*n*-Octane	8.163	70.567	−22.208			298	1500	1

Sources:
1. J. M. Smith, H. C. Van Ness, and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 5th ed. (New York: McGraw-Hill, 1996).
2. P. J. Linstrom and W. G. Mallard, Eds., **NIST Chemistry WebBook, NIST Standard Reference Database Number 69,** June 2005, National Institute of Standards and Technology, Gaithersburg MD, 20899 (http://webbook.nist.gov/chemistry/fluid).

TABLE A.2.2 Heat Capacity of Ideal Gases: Inorganic Compounds

Formula	Name	A	$B \times 10^3$	$C \times 10^6$	$D \times 10^{-5}$	$E \times 10^9$	T_{min}	T_{max}	Source
	Air	3.355	0.575		−0.016		298	2000	1
BCl_3	Boron trichloride	4.245	16.539	−18.969	−0.176	8.031	298	700	2
		9.882	0.078	−0.018	−3.374	0.001	700	6000	2
B_2H_6	Diborane	−1.494	32.188	−18.314	0.361	3.988	298	1200	2
		19.440	1.351	−0.262	−44.224	0.018	1200	6000	2
Br_2	Bromine	4.493	0.056		−0.154		298	3000	1
CF_4	Carbon tetrafluoride	1.921	25.299	−22.789	−0.261	7.482	298	1000	2
		12.776	0.129	−0.027	−10.032	0.002	1000	6000	2
CO	Carbon monoxide	3.376	0.557		−0.031		298	2500	1
CO_2	Carbon dioxide	5.457	1.045		−1.157		298	2000	1
CS_2	Carbon disulfide	6.311	0.805		−0.906		298	1800	1
C_2F_6	Hexafluoroethane	8.389	27.106	−20.948	−1.751	5.671	298	1400	2
		21.284	0.123	−0.023	−13.447	0.001	1400	6000	2
Cl_2	Chlorine	4.442	0.089		−0.344		298	3000	1
H_2	Hydrogen	3.249	0.422		0.083		298	3000	1
HBr	Hydrogen bromide	3.815	−1.648	2.809	−0.035	−1.084	298	1100	2
		3.956	0.339	−0.057	−3.819	0.004	1100	6000	2
HCN	Hydrogen cyanide	4.736	1.359		−0.725		298	2500	1
HCl	Hydrogen chloride	3.156	0.623		0.151		298	2000	1
HF	Hydrogen fluoride	3.622	−0.390	0.345	−0.030	0.055	298	1000	2
		2.955	0.829	−0.150	−0.282	0.010	1000	6000	2
H_2O	Water	3.470	1.45		0.121		298	2000	1
H_2S	Hydrogen sulfide	3.931	1.49		−0.232		298	2300	1
N_2	Nitrogen	3.280	0.593		0.04		298	2000	1
NH_3	Ammonia	3.5778	3.02		−0.186		298	1800	1
N_2O	Nitrous oxide	5.328	1.24		−0.928		298	2000	1
NO	Nitric oxide	3.387	0.629		0.014		298	2000	1
NO_2	Nitrogen dioxide	4.982	1.195		−0.792		298	2000	1
N_2O_4	Dinitrogen tetroxide	11.660	2.257		−2.787		298	2000	1
O_2	Oxygen	3.639	0.506		−0.227		298	2000	1
PH_3	Phosphene	1.431	10.160	−4.576	0.348	0.685	298	1200	2
SF_6	Sulfur hexafluoride	7.085	30.736	−30.343	−1.935	10.676	298	1000	2
		18.901	0.058	−0.012	−9.959	0.001	1000	6000	2
SO_2	Sulfur dioxide	5.699	0.801		−1.015		298	2000	1
SO_3	Sulfur trioxide	8.06	1.056		−2.028		298	2000	1
$SiCl_4$	Tetrachlorosilane	12.700	0.255	−0.069	−1.744	0.006	298	6000	2
$SiClH_3$	Chlorosilane	2.977	14.807	−9.231	−0.432	2.242	298	1100	2
		11.954	0.572	−0.114	−17.303	0.008	1100	6000	2
$SiCl_2H_2$	Dichlorosilane	6.026	10.145	−6.014	−0.959	1.328	298	1500	2
		12.603	0.195	−0.035	−15.277	0.002	1500	6000	2
$SiCl_3H$	Trichlorosilane	7.732	10.262	−8.671	−0.908	2.818	298	1000	2
		12.552	0.253	−0.052	−7.179	0.004	1000	6000	2
SiF_4	Silicon tetrafluoride	5.170	19.158	−18.179	−0.514	6.207	298	1000	2
		12.903	0.057	−0.012	−6.482	0.001	1000	6000	2
SiH_4	Silane	0.729	16.835	−9.368	0.163	1.953	298	1300	2
		12.010	0.511	−0.097	−24.525	0.006	1300	6000	2
WF_6	Tungsten hexafluoride	18.137	0.730	−0.197	−3.690	0.017	1000	6000	2

Sources:

1. J. M. Smith, H. C. Van Ness, and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 5th ed. (New York: McGraw- Hill, 1996).
2. P. J. Linstrom and W. G. Mallard, Eds., **NIST Chemistry WebBook, NIST Standard Reference Database Number 69,** June 2005, National Institute of Standards and Technology, Gaithersburg MD, 20899 (http://webbook.nist.gov/chemistry/fluid).

TABLE A.2.3 Heat Capacity of Liquids and Solids

Formula	Name	Phase	A	$B \times 10^3$	$D \times 10^{-5}$	Source
CH_4O	Methanol	L, $\bar{c}_P$	9.815			2
C_2H_6O	Ethanol	L, $\bar{c}_P$	13.592			2
C_3H_6O	Acetone	L	11.184	13.375		2
C_5H_{12}	Pentane	L	18.691	5.254		3
C_6H_6	Benzene	L	16.310	0.000		2
C_6H_{14}	Hexane	L	23.695			2
Al	Aluminum	L	3.819			1
Al	Aluminum	S	2.486	1.490		1
Al_2O_3	Aluminum oxide	S	23.154			4
C	Graphite	S	2.063	0.514	−1.057	1
C	Diamond	S	0.782			4
Cu	Copper	L	3.950			4
Cu	Copper	S	2.723			1
Cu_2O	Cuprous oxide	S, alpha	7.498			1
CuO	Cupric oxide	S	4.666			1
Fe	Iron	S, alpha	2.104	2.979		1
Fe_3O_4	Iron oxide	S	11.012	24.260		1
GaAs	Gallium arsenide	S	5.438	0.730		1
Ni	Nickel	S	1.508	4.308	0.297	1
Si	Silicon	L	3.272			4
Si	Silicon	S	2.879	0.297	−0.498	1
SiO_2	Silicon dioxide	S	5.647	4.127	−1.359	1
$SiCl_3H$	Trichlorosilane	L, $\bar{c}_P$	15.678			2
$SiCl_4$	Tetrachlorosilane	L, $\bar{c}_P$	16.117			2
H_2O	Water	L, $\bar{c}_P$	9.069			2
H_2O	Water (ice)	S, $\bar{c}_P$	4.196			5
H_2SO_4	Sulfuric acid	L	16.731	1.875		3
HNO_3	Nitric acid	L, $\bar{c}_P$	13.315			2
NH_3	Ammonia	L	6.880	9.682		2

Sources:

1. O. Kubaschewski and C. B. Alcock, *Metallurgical Thermochemistry*, 5th ed. (New York: Peramon Press, 1979).
2. Milan Zabransky et al., *Heat Capacity of Liquids* (Washington, DC: American Chemical Society; Woodbury, NY: National Bureau of Standards, 1996).
3. Richard M. Felder and Ronald W. Rousseau, *Elementary Principles of Chemical Processes*, 3rd ed. (New York: Wiley, 2000).
4. M. W. Chase et al., *JANAF Themochemical Tables*, 4th ed. (Washington, DC: American Chemical Society; National Bureau of Standards, 1998).
5. K. Ranjevic, *Handbook of Thermodynamic Tables and Charts* (New York: McGraw-Hill, 1976).

▶ A.3 ENTHALPY AND GIBBS ENERGY OF FORMATION AT 298 K AND 1 BAR

TABLE A.3.1 Organic Compounds

Formula	Name	Phase	$\Delta h^o_{f,298}$ [kJ/mol]	$\Delta g^o_{f,298}$ [kJ/mol]	Source
CH_2O	Formaldehyde	G	−115.97	−109.99	1
CH_4	Methane	G	−74.81	−50.72	1
CH_4O	Methanol	L	−238.73	−166.34	1
CH_4O	Methanol	G	−200.66	−161.96	1
C_2H_2	Acetylene	G	226.88	209.24	1
C_2H_3N	Acetonitrile	L	53.17	98.93	1
C_2H_3N	Acetonitrile	G	87.92	105.67	1
C_2H_4	Ethylene	G	52.26	68.15	1

(*Continued*)

TABLE A.3.1 Continued

Formula	Name	Phase	$\Delta h^o_{f,298}$ [kJ/mol]	$\Delta g^o_{f,298}$ [kJ/mol]	Source
$C_2H_4Cl_2$	1,1-Dichloroethane	L	−160.86	−76.20	1
$C_2H_4Cl_2$	1,1-Dichloroethane	G	−130.00	−73.14	1
C_2H_4O	Acetaldehyde	G	−166.47	−133.39	1
C_2H_4O	Ethylene oxide	L	−77.46	−11.43	1
C_2H_4O	Ethylene oxide	G	−52.67	−13.10	1
$C_2H_4O_2$	Acetic acid	L	−484.41	−389.62	1
$C_2H_4O_2$	Acetic acid	G	−435.13	−376.94	1
C_2H_6	Ethane	G	−84.68	−32.84	1
C_2H_6O	Ethanol	L	−277.17	−174.25	1
C_2H_6O	Ethanol	G	−234.96	−168.39	1
C_3H_6	Propylene	G	20.43	62.76	1
C_3H_6O	Acetone	L	−248.28	−155.50	1
C_2H_6O	Acetone	G	−217.71	−153.15	1
C_3H_6O	Propylene oxide	L	−120.75	−26.75	1
C_3H_6O	Propylene oxide	G	−92.82	−25.79	1
C_3H_8	Propane	G	−103.85	−23.49	1
C_3H_8O	1-Propanol	L	−304.76	−170.78	1
C_3H_8O	1-Propanol	G	−257.70	−163.08	1
C_4H_6	1,3-Butadiene	L	85.41	149.68	1
C_4H_6	1,3-Butadiene	G	110.24	150.77	1
C_4H_8	1-Butene	G	−0.13	71.34	1
C_4H_8	*cis*-2-Butene	G	−6.99	65.90	1
C_4H_8	*trans*-2-Butene	G	−11.18	63.01	1
$C_4H_8O_2$	Ethyl acetate	L	−479.35	−332.93	1
$C_4H_8O_2$	Ethyl acetate	G	−443.21	−327.62	1
C_4H_{10}	*n*-Butane	L	−147.75	−15.07	1
C_4H_{10}	*n*-Butane	G	−126.23	−17.17	1
C_4H_{10}	Isobutane	L	−158.55	−21.98	1
C_4H_{10}	Isobutane	G	−134.61	−20.89	1
$C_4H_{10}O$	*n*-Butanol	L	−326.03	−161.19	1
$C_4H_{10}O$	*n*-Butanol	G	−274.61	−150.77	1
C_5H_{10}	1-Pentene	L	−46.72	78.25	1
C_5H_{10}	1-Pentene	G	−20.93	79.17	1
C_5H_{12}	*n*-Pentane	L	−173.33	−9.46	1
C_5H_{12}	*n*-Pentane	G	−146.54	−8.37	1
C_6H_6	Benzene	L	49.07	124.34	1
C_6H_6	Benzene	G	82.98	129.75	1
C_6H_6O	Phenol	S	−165.13	−50.45	1
C_6H_6O	Phenol	G	−96.42	−32.91	1
C_6H_7N	Aniline	L	31.11	149.18	1
C_6H_7N	Aniline	G	86.92	166.80	1
C_6H_{12}	Cyclohexane	L	−156.34	26.89	1
C_6H_{12}	Cyclohexane	G	−123.22	31.78	1
C_6H_{12}	1-Hexene	L	−72.43	83.44	1
C_6H_{12}	1-Hexene	G	−41.70	87.50	1
C_6H_{14}	*n*-Hexane	L	−198.96	−4.35	1
C_6H_{14}	*n*-Hexane	G	−167.30	−0.25	1
C_7H_8	Toluene	L	12.02	113.84	1
C_7H_8	Toluene	G	50.03	122.09	1
C_7H_{14}	1-Heptene	L	−98.01	88.84	1
C_7H_{14}	1-Heptene	G	−62.34	95.88	1
C_7H_{16}	*n*-Heptane	L	−224.54	1.00	1

TABLE A.3.1 Continued

Formula	Name	Phase	$\Delta h^o_{f,298}$ [kJ/mol]	$\Delta g^o_{f,298}$ [kJ/mol]	Source
C_7H_{16}	*n*-Heptane	G	−187.90	8.00	1
C_8H_{10}	*o*-Xylene	L	−24.45	110.53	1
C_8H_{10}	*o*-Xylene	G	19.01	122.17	1
C_8H_{10}	*m*-Xylene	L	−25.41	107.73	1
C_8H_{10}	*m*-Xylene	G	17.25	118.95	1
C_8H_{10}	*p*-Xylene	L	−24.45	110.03	1
C_8H_{10}	*p*-Xylene	G	17.96	121.21	1
C_8H_{10}	Ethylbenzene	L	−12.48	119.78	1
C_8H_{10}	Ethylbenzene	G	29.81	130.67	1
C_8H_{16}	1-Octene	L	−123.59	94.16	1
C_8H_{16}	1-Octene	G	−82.98	104.29	1
C_8H_{18}	*n*-Octane	L	−250.12	6.49	1
C_8H_{18}	*n*-Octane	G	−208.59	16.41	1
C_9H_{20}	*n*-Nonane	L	−275.66	11.76	1
C_9H_{20}	*n*-Nonane	G	−229.19	24.83	1
$C_{10}H_8$	Naphthalene	S	78.13	201.18	1
$C_{10}H_8$	Naphthalene	G	151.06	223.74	1
$C_{10}H_{22}$	*n*-Decane	L	−301.24	17.25	1
$C_{10}H_{22}$	*n*-Decane	G	−249.83	33.24	1

Sources:
1. Daniel R. Stull, Edgar F. Westrum, and Gerard C. Sinke, *The Chemical Thermodynamics of Organic Compounds*, (New York: Wiley, 1969).

TABLE A.3.2 Inorganic Compounds

Formula	Name	Phase	$\Delta h^o_{f,298}$ [kJ/mol]	$\Delta g^o_{f,298}$ [kJ/mol]	Source
BCl_3	Boron trichloride	G	−402.96	−387.96	2
B_2H_6	Diborane	G	35.61	86.77	2
BN	Boron nitride	S	−254.387	−228.501	2
B_2O_3	Boron oxide	S	−1271.94	−1192.8	2
CCl_3F	Trichlorofluoromethane	G	−284.70	−245.51	1
CF_4	Carbon tetrafluoride	G	−975.52	−889.03	1
C_2F_6	Hexafluoroethane	G	−1343.06	−1257.3	2
$CHCl_3$	Chloroform	L	−132.30	−71.89	1
$CHCl_3$	Chloroform	G	−101.32	−68.58	1
CHN	Hydrogen cyanide	G	130.63	120.20	1
CO	Carbon monoxide	G	−110.53	−137.17	2
CO_2	Carbon dioxide	G	−393.51	−394.36	2
CS_2	Carbon disulfide	G	116.94	66.82	2
CaS	Calcium sulfide	S	−473.2	−468.178	2
$CaSO_4$	Calcium sulfate	S	−1434.11	−1321.68	2
CaO	Calcium oxide	S	−635.09	−603.51	2
CuCl	Copper chloride	S	−155.65	−138.66	2
CuO	Copper monoxide	S	−156.06	−128.29	2
Cu_2O	Dicopper oxide	S	−170.71	−147.88	2
CuS	Copper sulfide	S	−53.1	−53.47	2
$CuSO_4$	Copper sulfate	S	−771.36	−662.08	2
Fe_3C	Triiron carbide	S	25.104	20.029	2

(*Continued*)

TABLE A.3.2 Continued

Formula	Name	Phase	$\Delta h^o_{f,298}$ [kJ/mol]	$\Delta g^o_{f,298}$ [kJ/mol]	Source
Fe_2O_3	Hematite	S	−824.25	−742.29	2
Fe_2O_4	Magnetite	S	−1118.38	−1015.23	2
HBr	Hydrogen bromide	G	−36.38	−53.45	2
HCl	Hydrogen chloride	G	−92.312	−95.29	2
HF	Hydrogen fluoride	G	−272.55	−274.65	2
HNO_3	Nitric acid	G	−134.31	−73.96	2
H_2O	Water	L	−285.83	−237.14	2
H_2O	Water	G	−241.82	−228.57	2
H_2S	Hydrogen sulfide	G	−20.5	−33.33	2
H_2SO_4	Sulfuric acid	L	−813.99	−689.89	2
H_2SO_4	Sulfuric acid	G	−735.13	−653.37	2
NH_3	Ammonia	G	−46.11	−16.45	2
N_2O	Nitrous oxide	G	82.05	104.17	2
NO	Nitric oxide	G	90.29	86.6	2
NO_2	Nitrogen dioxide	G	33.1	51.26	2
N_2O_4	Dinitrogen tetraoxide	L	−19.564	97.51	2
N_2O_4	Dinitrogen tetraoxide	G	9.079	97.79	2
NaCl	Sodium chloride	S	−411.12	−384.02	2
NaF	Sodium fluoride	S	−573.48	−545.08	2
NaOH	Sodium hydroxide	S	−425.93	−379.73	2
NaOH	Sodium hydroxide	G	−197.49	−200.19	2
O	Oxygen	G	249.17	231.74	2
PH_3	Phosphene	G	5.57	13.59	2
TaN	Tantalum nitride	S	−252.3	−226.58	2
TiC	Titanium carbide	S	−184.5	−180.84	2
TiN	Titanium nitride	S	−337.86	−309.16	2
SiC	Silicon carbide	S	−73.22	−70.85	2
$SiCl_2$	Dichlorosilylene	G	−168.61	−180.36	2
$SiCl_4$	Silicon tetrachloride	G	−662.75	−622.76	2
SiF_4	Silicon tetrafluoride	G	−1614.94	−1572.7	2
$SiCl_3H$	Trichlorosilane	G	−496.22	−464.9	3
$SiCl_2H_2$	Dichlorosilane	G	−320.49	−294.9	3
SiH_3Cl	Chlorosilane	G	−141.838	−119.29	3
SiH_4	Silane	G	34.31	56.82	2
Si_3N_4	Silicon nitride	S	−744.75	−647.34	2
SiO_2	Silicon dioxide, trigonal	S	−910.86	−856.44	2
SiO_2	Silicon dioxide, hexagonal	S	−906.34	−757.11	2
SiO_2	Silicon dioxide, cristobalite	S	−902.53	−716.46	2
SiO_2	Silicon dioxide	L	−935.34	−551.67	2
SF_6	Sulfur hexafluoride	G	−1220.47	−1116.5	2
SO_2	Sulfur dioxide	G	−296.813	−300.1	2
SO_3	Sulfur trioxide	G	−395.77	−371.02	2
WF_6	Tungsten hexafluoride	G	−1721.72	−1632.29	2
ZnO	Zinc oxide	S	−350.46	−320.48	2
ZnS	Zinc sulfide, wurtzite	S	−191.84	−190.14	2
ZnS	Zinc sulfide, sphalerite	S	−205.18	−200.4	2
$ZnSO_4$	Zinc sulfate	S	−982.8	−871.45	2

Sources:

1. Daniel R. Stull, Edgar F. Westrum, and Gerard C. Sinke, *The Chemical Thermodynamics of Organic Compounds* (New York: Wiley, 1969).
2. Ihsan Barin, *Thermochemical Data of Pure Substances*, 3rd ed. (vol. I and II) (New York: VCH, 1995).
3. M. W. Chase et al., *JANAF Thermochemical Tables*, 3rd ed. (Washington, DC: American Chemical Society; (New York: National Bureau of Standards, 1986).